

Thermal Desorption Spectroscopy

Group C

Zhan Yi Bill Chng, zchng@smail.uni-koeln.de

Alexander Bott, s6albott@uni-bonn.de

Supervisor: Anna Kny

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In this experiment, the desorption of phenol from a clean Ag(111) surface was studied. Residual gas spectra in addition to mass spectra of benzene and phenol were recorded before applying phenol onto the clean surface. TPD-spectra with different coverages of phenol on the surface were recorded and characterised as monolayer or multilayer desorption. From the data, the sticking factor of phenol on an Ag(111) surface and the desorption energies for monolayer and multilayer desorption were calculated.

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1 Introduction

Thermal desorption spectroscopy (TDS), also known as temperature programmed desorption (TPD), is a common means of studying the kinetics of desorption processes on surfaces. This provides quantitative information about the binding energy of molecules to a surface.

Different dosages of a gas can be applied onto the cleaned and annealed Ag(111) surface in an evacuated chamber for adsorption. As temperature is raised the adsorbed molecules start to desorb of the surface. In general, most desorption processes can be distinguished in three distinct categories. First order desorption describes molecules adsorbed on a monolayer, desorbing as a complete molecule. For example, a physisorbed molecule that does not undergo any chemical change typically obeys first order desorption. In such cases, the rate of desorption is proportional to the coverage.

In second order desorption, the adsorbed molecules combine together on the surface prior to desorption, such that this combination step is rate-limiting. This necessitates a chemical change in the adsorbates. The bimolecularity of this rate-determining step results in the overall rate of desorption being dependent on the square of the coverage. The zero order desorption describes any desorption of molecules adsorbed not directly on the surface but on another layer of adsorbates. This desorption of the molecules takes place at set temperatures, which are proportional to the desorption energy E_{des} . TPD-measurements are then calibrated on a set mass per charge ratio and observe the desorption of molecules with said mass per charge ratio at a certain temperature. In general, the rate of desorption, r_{des} , can be modelled *via* the Polanyi-Wigner equation (Equation 1). If the temperature of desorption is independent of the amount of adsorbate, the adsorbate obeys first order desorption ($m = 1$).

$$r_{des} = \nu_0 \theta_0^m \exp\left(-\frac{E_{des}}{k_B T}\right) \quad (1)$$

where ν_0 is the pre-exponential factor, θ_0 is the initial coverage, and m is the order of desorption.

This trend changes for dosages that lead to coverages higher than one monolayer. Because of their distance and the shielding by the first layer adsorbates, the adsorbates in the higher layer have a much weaker bond onto the surface than the adsorbates of the first layer. Consequently, the energy which is needed to desorb these molecules from the surface is normally smaller. This difference can even be seen for the second layer of adsorbates in the TPD-spectra as a new desorption peak at a lower temperature.

If the dosage is increased, the peak of the second layer molecules will saturate when the second layer of adsorbates is filled. Further increasing the dosage leads to a third peak forming at slightly lower temperature, which indicates the desorption of molecules of the third or higher layers. These layers cannot be distinguished anymore in their desorption energies, because their desorption energy difference is too small. This results in a much bigger peak compared to the ones corresponding to the first and second layers at high coverages. With this principle, TDS also gives information about the coverage of an adsorbate on the surface and so about the sticking factor of the molecules on the surface.^[1,2]

2 Methodology

2.1 Experimental setup

The temperature programmed desorption was carried out in a vacuum chamber kept at ultra high vacuum conditions (UHV), *i.e.*, at a pressure of *ca.* $1 \cdot 10^{-9}$ mbar. The UHV was maintained *via* a Pfeiffer Vacuum DN-100 CF-F Turbopump, with a nominal pumping speed of 260 L s^{-1} . The pressure was measured with an ionisation gauge manometer. Gaseous phenol or benzene were dosed into the chamber through leak valves, which leak rate was indirectly monitored *via* the pressure reading on the manometer. The sample was mounted on a holder attached to a manipulator that allows for 260° of rotation. The sample can be cooled to 73 K with a liquid nitrogen cryostat or heated with a tungsten filament to 700 K. Connected to the vacuum chamber was a Quadrupole Mass Spectrometer (QMS), used to analyse and monitor the species being desorbed.^[1]

2.2 Experimental procedure

2.2.1 Residual gas and adsorbent analysis

The QMS was initialised with the following settings: Range = 10^{-9} A, Gain = 1, Speed = $1 \frac{m}{z} \text{ s}^{-1}$, First Mass = 0, Width ($\frac{m}{z}$) = 100. A residual gas spectrum was measured from the evacuated chamber ($p = 2.9 \cdot 10^{-9}$ mbar) with using the inbuilt QMS. After that, the filament of the ionisation gauge manometer was degassed ($p_{max} = 2.8 \cdot 10^{-8}$ mbar), and a mass spectrum was obtained simultaneously. Note that the Range was changed to 10^{-7} on the QMS just for this measurement. Once the degassing was completed, Argon was filled into the chamber until a pressure of $1 \cdot 10^{-8}$ mbar was reached, and another mass spectrum was recorded. Before continuation of the measurement, the pressure was again lowered to $1 \cdot 10^{-9}$ mbar. Phenol was then dosed into the vacuum chamber until a pressure of $1 \cdot 10^{-8}$ mbar, and a mass spectrum was recorded. The molecular and residual gas peaks were determined. This was repeated for benzene.^[1]

2.2.2 Temperature programmed desorption experiments

In this study, the sample used was an Ag(111) surface. To ensure that the surface is clean, it was first sputtered with Ar^+ with an argon-ion gun ($p_{Ar} = 1 \cdot 10^{-5}$ mbar, Acceleration voltage = 1 kV, Emission current = 20 mA, Sample current = $0.7519 \mu\text{A}$) for 20 minutes. The sample was then annealed ($T_{max} = 700 \text{ K}$, $\beta = 1.00 \text{ K s}^{-1}$) for 40 minutes, and then left to cool to 100 K. The adsorbent chosen for the TPD experiments was phenol. Each TPD experiment starts with dosing phenol into the chamber at a constant pressure until the target dosage was reached, ensuring that the sample is aligned towards the phenol leak valve. The sample was then rotated to the QMS port, and the QMS was set to select for a mass of 94 u., corresponding to the molecular peak for phenol ($\frac{m}{z}[\text{M}^+] = 93.0$, offset by 1 unit of $\frac{m}{z}$ in QMS readings). A TPD-spectrum was then recorded from $T = 120 \text{ K}$ to $T = 350 \text{ K}$. This was repeated for different dosages of phenol ranging from $3.0 \cdot 10^{-8} \text{ mbar} \cdot \text{s}$ to $7.0 \cdot 10^{-7} \text{ mbar} \cdot \text{s}$.^[1]

3 Results

3.1 Residual gas analysis

3.1.1 Residual gas spectra of the empty chamber

To ascertain at which $\frac{m}{z}$ should the TPD-measurement take place, different residual gas spectra (RG-spectra) were recorded. A problem for this measurement are the few different molecules which remain, even in UHV conditions. Thus, the first RG-spectra which were recorded are one of the evacuated chamber, one where the chamber was degassed and one where argon is flooded into the chamber.

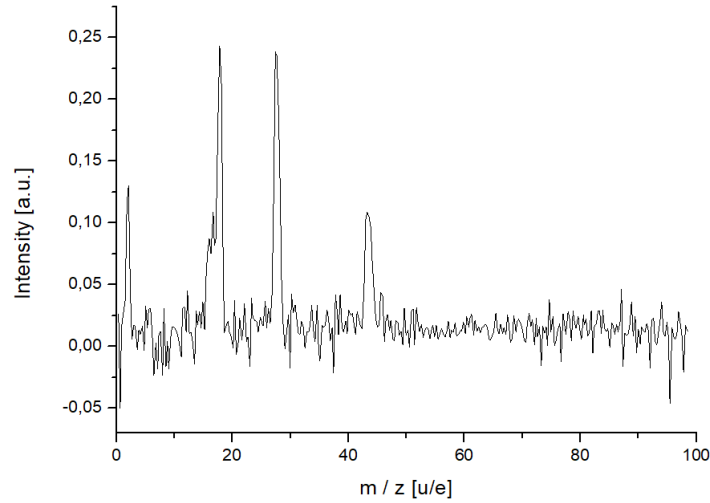


Figure 1: RG-spectra of the evacuated chamber at the start of the experiment at a pressure of $1.0 \cdot 10^{-9}$ mbar. The intensity is plotted against the mass divided by the charge of the molecule. Only the residual gas of the evacuated chamber is seen.

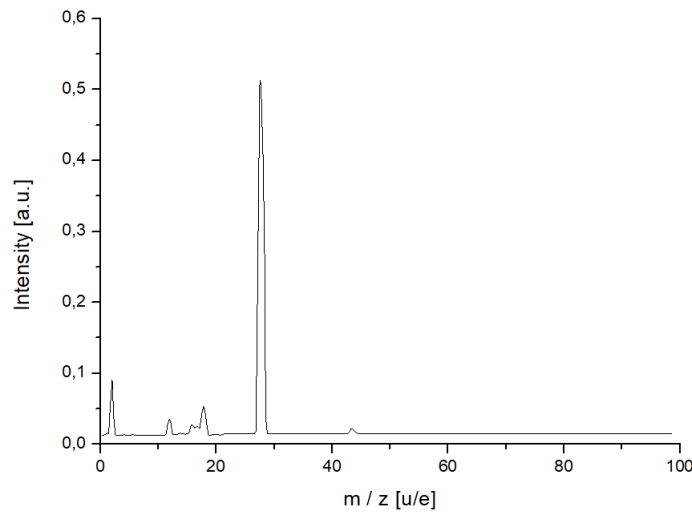


Figure 2: RG-spectra of the evacuated chamber at a pressure of $2.8 \cdot 10^{-8}$ mbar while degassing the filament. The intensity is plotted against the mass divided by the charge of the molecule.

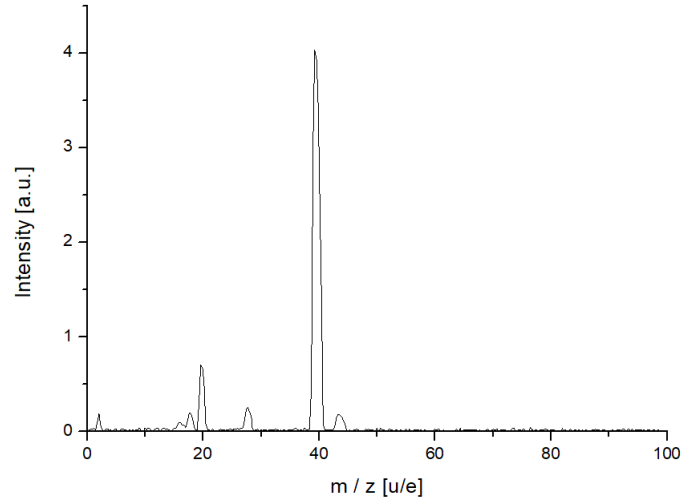


Figure 3: RG-spectra of the evacuated chamber flooded with argon at a pressure of $1 \cdot 10^{-8}$ mbar. The intensity is plotted against the mass divided by the charge of the molecule.

Table 1: Summary and appointment of the RG-spectra in Figure 1, 2 and 3.

m/z [u/e]	Molecule
2	H_2^+
12	C^+
16, 17, 18	O^+ , OH^+ , H_2O^+
28	CO^+ , N_2^+
20, 40	Ar^{2+} , Ar^+
44	CO_2^+

The specific appointment to the peaks is visible in table 1. While CO_2 and N_2 are residual of the normal atmosphere, CO and C are mainly from dissociation at hot filaments and leftover peaks of separated CO_2 and Argon is used for sputtering the surface, the high intensity for water is explainable by the high tendency of H_2O to adsorb on surfaces, as for example at the metal wall of the chamber. Because these particles, at least in some degree, will always appear in our RG-spectra, in the following RG-spectra they can be neglected.

3.1.2 Residual gas spectra of phenol and benzene

To find out the molecular ion peak for benzene and phenol, which will be used subsequently for TPD experiments, a mass spectrum was recorded for each of the compounds at a pressure of $1 \cdot 10^{-8}$ mbar.

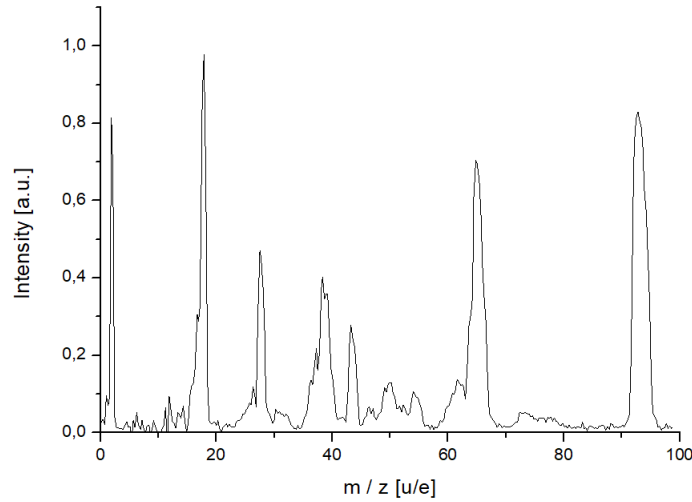


Figure 4: Mass spectrum of phenol at a pressure of $1 \cdot 10^{-8}$ mbar. The intensity is plotted against the mass divided by the charge of the molecule.

Table 2: List of all phenol peaks out of Figure 4, specified with their intensity. The signals from the residual gas are neglected.

m/z [u/e]	intensity	molecule
38	0.41	C_3H_3^+
50	0.13	C_4H_3^+
54	0.11	$\text{C}_3\text{H}_2\text{O}^+$
62	0.14	C_5H_2^+
65	0.71	C_5H_6^+
93	0.83	$\text{C}_6\text{H}_5\text{O}^+$

The first of the two mass spectra was the one for Phenol, flooded to a pressure of $1 \cdot 10^{-8}$ mbar. There are six additional peaks in the spectra in Figure 4 in comparison to those peaks listed in Table 1, which all can be sorted to phenol and are shown in Table 2. The most intense peak is the one at 93 u/e, which is close to the calculated molecular mass of 94 u, which was also used in the publication of J. Lee et al.^[3]. A reason for this difference is that the alcohol-group on phenol is deprotonated. This is underlined by the small peak at 1 u/e, which can be assigned to H^+ . The other peaks for phenol can be described by the fragmentation of phenol during the mass spectroscopy, but they are all lower in intensity than the main peak.

So the most suitable mass for measuring a phenol TPD-spectra would be 93 u/e.

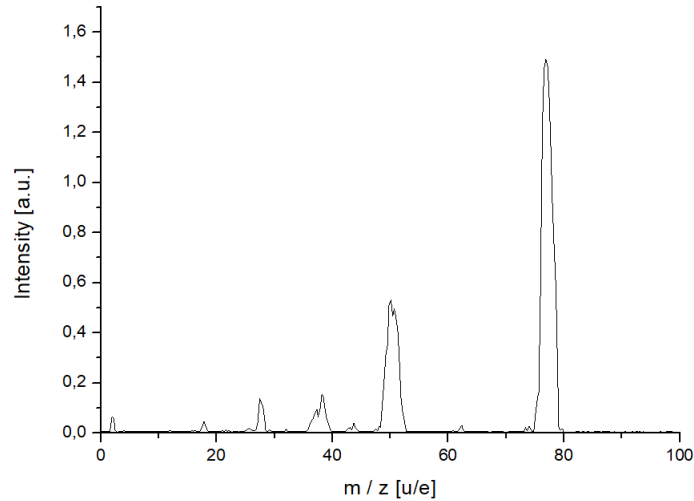


Figure 5: Mass spectrum of benzene at a pressure of $1 \cdot 10^{-8}$ mbar. The intensity is plotted against the mass divided by the charge of the molecule.

Table 3: List of all benzene peaks out of Figure 5, specified with their intensity. The signals from the residual gas are neglected.

m/z [u/e]	intensity	molecules
28	0.15	C_2H_4^+
37	0.1	C_3H_2^+
38	0.16	C_3H_3^+
50	0.53	C_4H_3^+
77	1.49	C_6H_5^+

The mass spectrum of benzene is shown in figure 5 and summarized in table 3. Its highest intensity was measured at a mass of $\frac{m}{z} = 77$ u/e. It stands out that again this peak differs by 1 u to the calculated molecular mass of benzene of 78 u. This could again be explained with a deprotonation, which should be much more unlikely in benzene than in phenol. The other peaks in the spectra are again attributed to the fragmentation of the benzene spectra. The mass, which is most suited to measure benzene, would be 77 u/e.

3.2 TPD-spectra

For better comparison, the TPD-spectra of different phenol dosages were all plotted in one diagram.

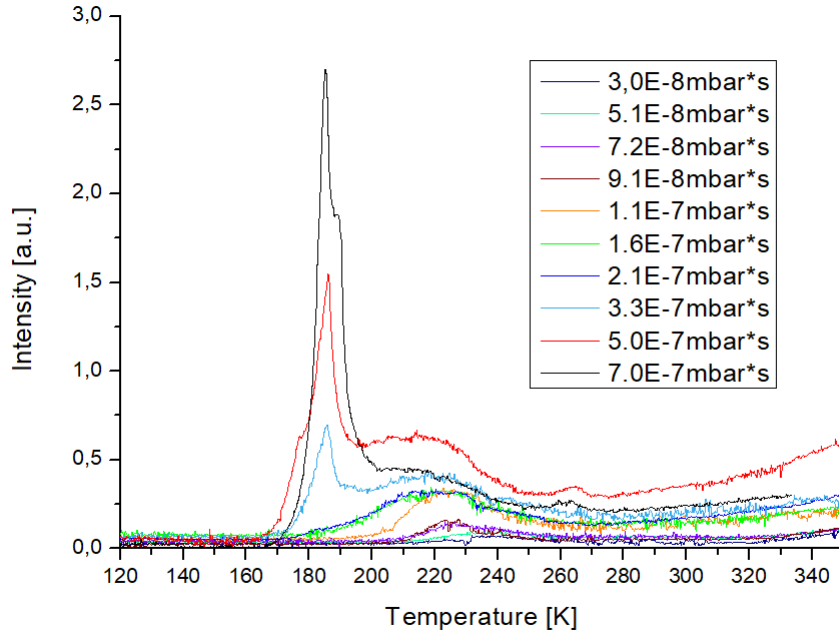


Figure 6: TPD-spectra of phenol adsorbed on a Ag(111) surface at 100-130 K. The heating rate was 1 K s^{-1} . The intensity is plotted against the temperature in K for ten different dosages ranging from $3.0 \cdot 10^{-8} \text{ mbar} \cdot \text{s}$ to $7.0 \cdot 10^{-7} \text{ mbar} \cdot \text{s}$.

In Figure 6 shown are the measured TPD-spectra, ranging from the lower dosage of $3.0 \cdot 10^{-8} \text{ mbar} \cdot \text{s}$ to the higher dosage of $7.0 \cdot 10^{-7} \text{ mbar} \cdot \text{s}$. There is a clear difference between the lower and the higher dosages visible. The spectrum can be split into three peaks.

The first, rather broad, peak can be seen in the spectra of all the dosages. With higher dosage, the peak rises and shifts into a lower temperature region, starting at 234 K and shifting to 214 K.

With its form and behaviour it is safe to assume, that this peak describes the monolayer desorption. At lower dosage, and so lower coverage, the molecules only form a monolayer on the surface. The desorption of named monolayer results in one broad peak. Compared to the experiments of J. Lee et al.^[3], the desorption has a similar form and lays in roughly the same temperature region of 230-250 K. Only the shift into lower temperature is untypical for first order desorption, which was also recorded in their data. J. Lee et al.^[3] came in their experiment to the conclusion, that the desorption kinetics of phenol do not strictly follow first order desorption but rather first order desorption with considerable lateral repulsive interaction between the adsorbates. This repulsion between the adsorbates leads to a smaller needed desorption energy and so to a lower desorption temperature. The more adsorbates there are on the surface, the higher is the repulsion between them and the lower the desorption temperature gets. It seems that this spectrum supports this conclusion.

The other peak is not visible in all spectra but only in these with a higher dosage. It is located at a temperature of 186 K and is first seen in really small intensity at a dosage of $2.1 \cdot 10^{-7}$ mbar·s. At higher dosage, it forms a sharp peak of high intensity and shifts about 4 K into higher temperature. The energy level and form of the peak lead to the conclusion, that we have molecules adsorbed on the surface with other surroundings than the phenol in the monolayer. The peak describes the doublelayer desorption, so the desorption of zero order. The phenol adsorbed in the second layer has a lower bonding energy to the surface and is so easier to desorb.

At the higher dosage of $5.0 \cdot 10^{-7}$ mbar·s, the peak saturates at a temperature of 186 K and gets a shoulder peak at a temperature of 177 K, which rises above the height of the doublelayer-desorption peak at a dosage of $7.0 \cdot 10^{-7}$ mbar·s. This indicates the existence of more than a third layer of phenol on the surface with even less bonding energy to the surface than the second layer. This is also described with zero order desorption but of the multilayer. Compared to the data of J. Lee et al.^[3], the doublelayer and multilayer desorption at higher coverages of this experiment is located at roughly 20 K smaller temperature than in the literature, but the form and course of the peak with dosage is similar to their data. This is close enough to confirm the doublelayer and multilayer desorption.

It is needed to mention the rise of desorption in even higher temperatures. This rise in temperature is not mentioned in the publication from J. Lee et al.. Since in the present work the cooling system did not work perfectly and even malfunctioned once, it can be said that the mentioned rise in desorption came from at the sample holder adsorbed phenol molecules, which desorb when the sample holder starts to get warm. This desorption will not be included in the following data analysis, ranging from 270-350 K.

The data of an additional data-set for a dosage of $2.6 \cdot 10^{-7}$ mbar·s is not included because of an error in the associated data-set.

3.2.1 The ideal monolayer

To perform an analysis involving the coverage, the TPD-spectra being closest to the desorption of the ideal monolayer must be identified. Three of the TPD-spectra were plotted for this reason.

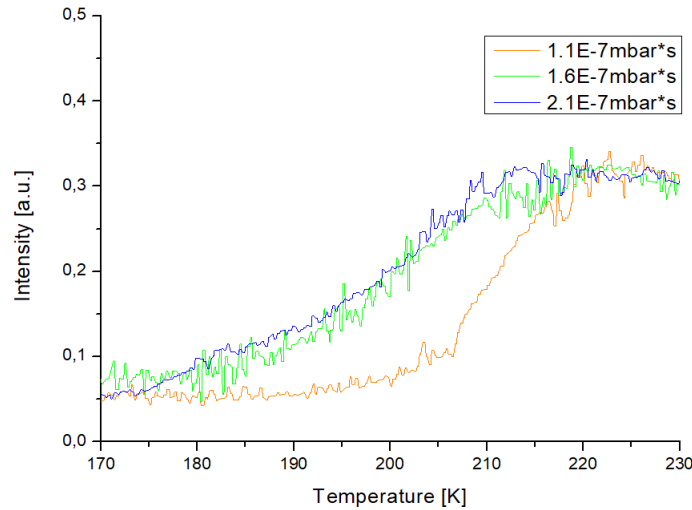


Figure 7: TPD-spectra of phenol adsorbed on a Ag(111) surface at 100-130K. The heating rate was 1 K s^{-1} . The intensity is plotted against the temperature in K for the three dosages of $1.1 \cdot 10^{-7} \text{ mbar} \cdot \text{s}$, $1.6 \cdot 10^{-7} \text{ mbar} \cdot \text{s}$, and $2.1 \cdot 10^{-7} \text{ mbar} \cdot \text{s}$. For detailed analysis, only the temperature region between 170 K and 230 K is displayed.

The three TPD-spectra of the dosages $1.1 \cdot 10^{-7} \text{ mbar} \cdot \text{s}$, $1.6 \cdot 10^{-7} \text{ mbar} \cdot \text{s}$, and $2.2 \cdot 10^{-7} \text{ mbar} \cdot \text{s}$ are the ones closest to an ideal monolayer. While the desorption of the multilayer is clearly visible in the desorption-spectrum of the $2.2 \cdot 10^{-7} \text{ mbar} \cdot \text{s}$ dosage, the difference between the other two dosages is only the width. The form of the monolayer desorption peak of the $1.6 \cdot 10^{-7} \text{ mbar} \cdot \text{s}$ dosage is much closer to that of the $2.2 \cdot 10^{-7} \text{ mbar} \cdot \text{s}$ dosage. The peak of the $1.1 \cdot 10^{-7} \text{ mbar} \cdot \text{s}$ dosage desorption-spectrum starts to rise at a higher energy level of 206 K instead of the 180 K by the other two dosages. One more small difference between the spectrum at the dosage of $1.1 \cdot 10^{-7} \text{ mbar} \cdot \text{s}$ and the other two is the small shift of the desorption maxima from 224 K at the dosage of $1.1 \cdot 10^{-7} \text{ mbar} \cdot \text{s}$ to 219 K at the other two.

Because its monolayer desorption is a little broader than the one at $1.1 \cdot 10^{-7} \text{ mbar} \cdot \text{s}$, but shows no signs of doublelayer desorption, it was decided that the dosage closest to the ideal monolayer is $1.6 \cdot 10^{-7} \text{ mbar} \cdot \text{s}$. The broad shape of the 1 ML peak is expected, because phenol in higher coverages binds to several different binding sites, which all have a small difference in desorption energy. The difference in desorption energy is in fact so small, that the different peaks of these binding sites are practically not distinguishable. Visible is only one combined broad peak of the different positioned phenol.

3.2.2 Initial coverages

The intensity of the signal produced by phenol at $\frac{m}{z} = 94$, $I(T)$ (in arbitrary units, a.u.), can be converted to the rate of desorption, r_{des} , via the following equation:

$$r_{des} = f_r \beta I(T) . \quad (2)$$

f_r here is termed the response factor, with dimensions of ML (a.u.)⁻¹ K⁻¹, whereas β is the heating rate, which in this experiment is 1 K s⁻¹. f_r can be calculated from the desorption peak integrals of the recorded spectra, as follows:

$$\int_{T_1}^{T_2} r_{des}(T) dT = f_r \beta \int_{T_1}^{T_2} I(T) dT . \quad (3)$$

Given that the integral for r_{des} with respect to temperature is just $\beta \cdot \theta_0$, this leads to,

$$f_r = \frac{\theta_0}{\int_{T_1}^{T_2} I(T) dT} . \quad (4)$$

The integrals of the desorption peak corresponding to the TPD-spectra for each dosage, D , were thus calculated, as shown in Table 4.

Table 4: Integrals from T_1 to T_2 for each dosage of phenol, D , and their corresponding initial coverages, θ_0

D [10^{-8} mbar s]	T_1 [K]	T_2 [K]	Integrals [(a.u.) K]	θ_0 [ML]
3.0	187.5	275.0	1.375	0.151
5.1	205.1	266.1	1.084	0.119
7.2	201.0	273.8	2.437	0.267
9.1	196.0	272.7	2.739	0.300
11.0	187.4	282.4	8.088	0.885
16.0	177.8	285.1	9.135	1.000
21.0	168.6	274.3	10.524	1.152
33.0	166.0	276.0	20.349	2.228
50.0	161.9	256.7	33.888	3.710
70.0	163.6	251.8	36.936	4.043

Using Equation 4, f_r was calculated with the integral for $D = 16 \cdot 10^{-8}$ mbar s, which was determined prior to correspond to 1 ML. The response factor was therefore ascertained to be 0.10946 ML (a.u.)⁻¹ K⁻¹. This value was used to determine the coverages for all other dosages, and the results were plotted against dosage. A linear regression ($R^2 = 0.97$) was obtained (Figure 8): $\theta = -0.07793 + 0.06492 \cdot D$.

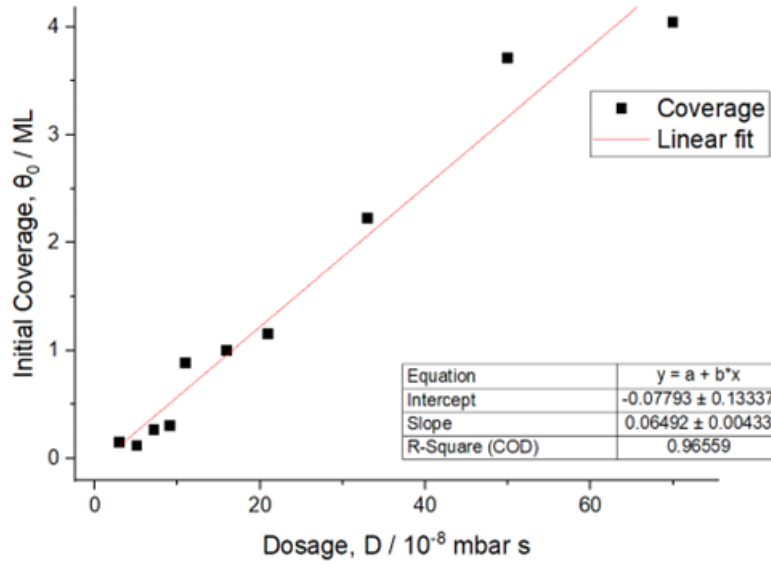


Figure 8: Diagram of the initial coverage θ_0 in ML against the dosage of the different TPD spectra in mbar.s. It was added a linear fit through the data with its data in the bottom right box.

3.2.3 Estimating the sticking factor

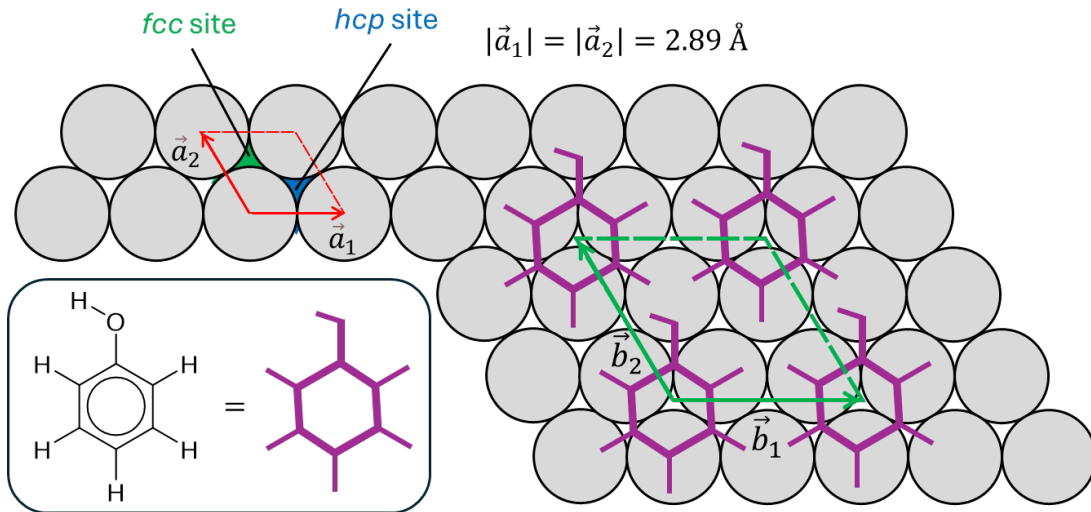


Figure 9: An ideal Ag(111) surface, with the surface unit cell illustrated. The overlayer unit cell shown above corresponds to $p(2 \times 2)$. The phenol molecules drawn above are dimensionally accurate relative to the Ag(111) surface. The bond lengths for phenol were obtained from the XRD reported by Zavodnik *et al.*^[4]

In the following analysis, the number of molecules in the monolayer was first estimated. As seen below in Table 5, impingement, N_{imp} , has to be converted from molecules cm^{-2} to in terms of monolayers, thus the number of molecules in the monolayer per cm^2 is needed. This can be calculated by examining an ideal (111) surface of an *fcc* lattice (Figure 9), with a surface lattice constant, $a = 2.89 \text{ \AA}$, as is the case for Ag (111). The area of the surface unit cell was determined to be $7.24 \text{ \AA}^2$. Each surface unit cell has one *fcc* and one *hcp* site, and it was assumed that

phenol cannot occupy both of them within the same surface unit cell. This is because the surface area covered by a phenol molecule (*ca.* $15 \text{ \AA}^2$) is much larger than that of the surface unit cell. For most (111) surfaces, *fcc* and *hcp* hollow are not energetically equivalent, because unlike an *fcc* hollow, there is an atom directly underneath the *hcp* site, which exerts an influence on the adsorption energy depending on the type adsorbate.^[5] Consequently, the maximum surface density of adsorption sites, *e.g.*, one *fcc* site for every $7.24 \text{ \AA}^2$, was calculated to be $1.38 \cdot 10^{15} \text{ cm}^{-2}$.

From here, an assumption has to be made for the size of the surface unit cell of the adsorbed overlayer. Overlaying phenol on top of Ag(111) (Figure 9) yields a $p(2 \times 2)$ structure containing one phenol molecule, which makes it the densest possible overlayer with an overlayer surface density of $3.45 \cdot 10^{14} \text{ cm}^{-2}$. Unfortunately, this results in all sticking factor values being much higher than 1 (Table 5), a result that is physically inviable. Thus, the assumed surface unit cell must produce sticking factors of ≤ 1 . A less dense unit cell of $p(6 \times 6)$ with one phenol adsorbed per unit cell was assumed, which gave an overlayer surface density of $3.83 \cdot 10^{13} \text{ cm}^{-2}$. This assumption was necessary to produce physically valid values of sticking factor and similar to those reported by J. Lee et al. (0.56).^[3] Ultimately, this corresponded to a surface density for a phenol monolayer of $3.8347 \cdot 10^{13} \text{ molecules cm}^{-2}$.

The average collision rate $\langle Z \rangle$, as Phenol was dosed, was calculated using this expression of impingement rate as a function of average dosing pressure $\langle p \rangle$:

$$\langle Z \rangle = \frac{\langle p \rangle}{2\pi m k_B T_{imp}} , \quad (5)$$

T_{imp} is the temperature when Phenol was impinged, and m is the mass of phenol, which is $1.561 \cdot 10^{-25} \text{ u}$. Since $N_{imp} = \int_{t_1}^{t_2} Z(t) dt$, relates directly to the dosage, D , the number of molecules impinged was calculated using a variant of equation (1), shown below.

$$N_{imp} = \frac{D}{2\pi m k_B T_{imp}} . \quad (6)$$

The impingement was then expressed in terms of monolayers (ML), using the surface density of Phenol calculated earlier. The estimated sticking factor for each dosage was then computed by dividing initial coverage, by N_{imp} , which results were plotted. No consistent sticking factor was observed, and two of the values exceeded 1, which is physically unrealistic, as that indicates more molecules were adsorbed than impinged.

Table 5: Variables involved in calculating Sticking factor, S , for 10 different dosages. D , corresponding to different initial coverages, θ_0 .

D [10^{-8} mbar s]	θ_0 [ML]	T_{imp} [K]	N_{imp} [10^{13} cm $^{-2}$]	N_{imp} , p(2 $\times$ 2) [ML]	N_{imp} , p(6 $\times$ 6) [ML]	S , p(2 $\times$ 2)	S , p(6 $\times$ 6)
3.0	0.1506	100.8	0.812	0.0235	0.2118	6.397	0.711
5.1	0.1186	101.1	1.380	0.0400	0.3601	2.965	0.329
7.2	0.2667	101.1	1.950	0.0565	0.5083	4.722	0.525
9.1	0.2998	101.6	2.460	0.0714	0.6425	4.199	0.467
11.0	0.8853	105.3	2.980	0.0863	0.7766	10.260	1.140
16.0	1.0000	102.4	4.330	0.1255	1.1296	7.967	0.885
21.0	1.1520	103.5	5.690	0.1647	1.4826	6.993	0.777
33.0	2.2275	104.3	8.930	0.2589	2.3298	8.605	0.956
50.0	3.7095	103.8	13.500	0.3922	3.5301	9.457	1.051
70.0	4.0431	117.0	19.000	0.5491	4.9421	7.363	0.818

Another means of obtaining the average sticking factor would be to fit a linear regression over a plot of θ_0 against impingement, N_{imp} . It can be observed that the sticking factor differs between $\theta_0 \leq 1$ and $\theta_0 \geq 1$. In the monolayer regime, the mean sticking factor is 0.47, whereas in the multilayer regime the sticking factor is 0.87. This is similar to the findings reported in the Lee paper, where $S = 0.56$ for monolayer coverages, and $S = 1.0$ for multilayer coverages.^[3]

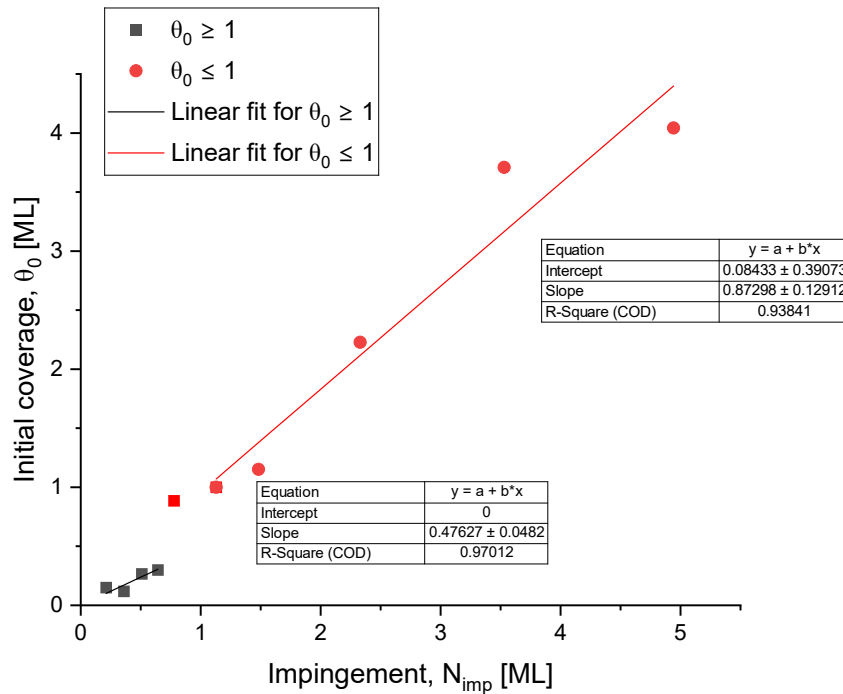


Figure 10: A plot of initial coverage plotted against impingement, the resultant slope of the linear regression is the sticking factor.

3.3 Multilayer desorption analysis

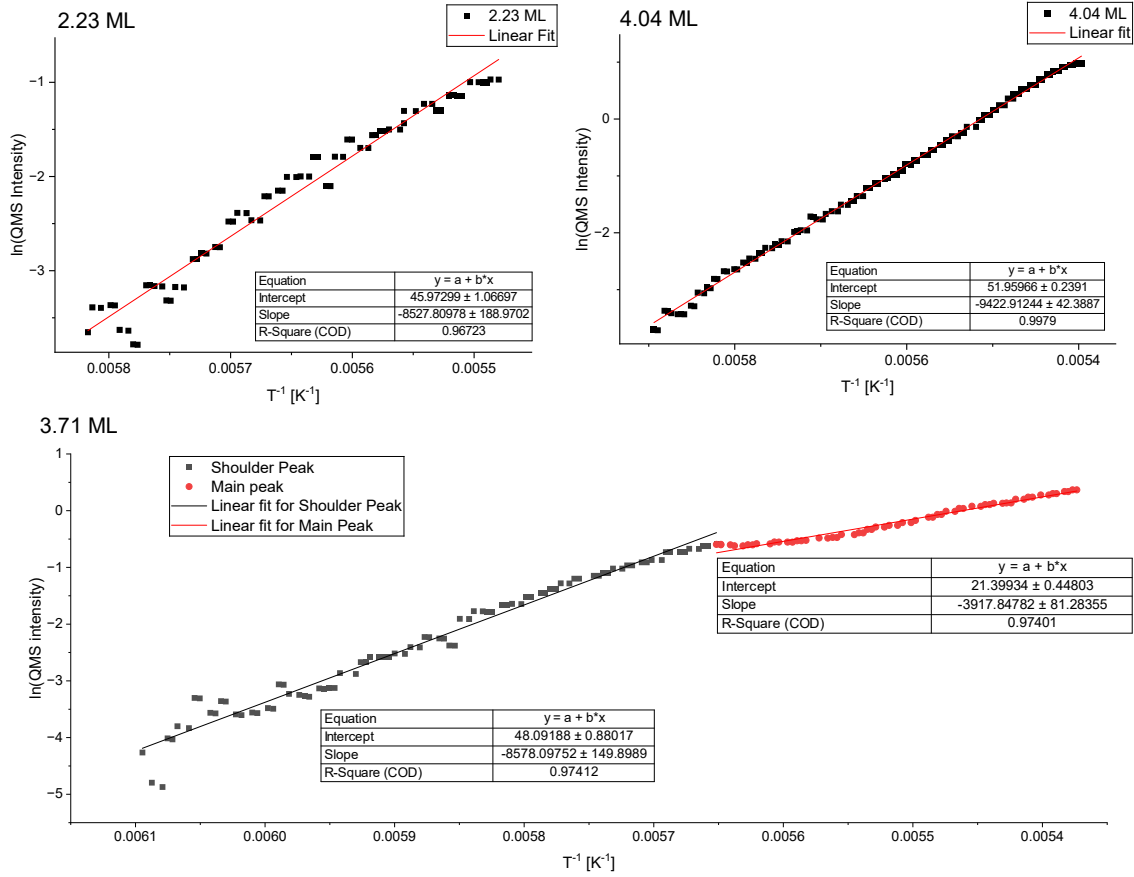


Figure 11: Three Arrhenius plots for different initial coverages [ML], where the $\ln(\text{QMS intensity})$ is plotted against $\frac{1}{T} [\text{K}^{-1}]$. Given are three different coverages, 2.23 ML, 3.71 ML and 4.04 ML, while the coverage of 3.71 ML has plots of two different peaks. All diagrams have a linear fit through their data, which properties are listed in the box in the bottom right of every linear fit.

As mentioned multilayer desorption is coverage independent, thus it obeys zeroth order desorption kinetics, as per Equation 1. This was linearised to obtain the following form:

$$\ln(r_{des}) = \left(-\frac{E_{des}}{k_B} \right) \frac{1}{T} + \ln(\nu_0), \quad (7)$$

with the slope $= -\frac{E_{des}}{k_B}$, and the y-intercept $= \ln(\nu_0)$.

Thus, Arrhenius plots of $\ln(r_{des})$ against $\frac{1}{T}$ were generated for each multilayer peak for TDS with coverages of higher than 1 ML (Figure 11). This yielded four linear plots from TDS with coverages of 2.23, 3.71, and 4.04 ML, with two plots from 3.71 ML as it has a shoulder peak at $T_p = 177 \text{ K}$ and a main peak at $T_p = 186 \text{ K}$. E_{des} was determined directly from their respective slopes, whereas ν_0 was calculated from the y-intercept (Table 6). From this, an E_{des} of $73.5 \pm 3. \text{ kJ mol}^{-1}$ was obtained. This value is in agreement with the multilayer desorption energy reported by J. Lee et al., which is $70.7 \pm 2.5 \text{ kJ mol}^{-1}$.^[3] The E_{des} for the main peak of 3.71 ML (38.5 kJ mol^{-1}) was excluded as the calculated value was significantly different from the other E_{des} values. This is most likely influenced by the significant shoulder

peak at 176.9 K. Unlike the desorption energy, the calculated ν_0 did not converge on a single value.

Table 6: Linear regression slope and intercept values for each peak at temperature, T_p , and initial coverage, θ_0 , used to compute E_{des} and ν_0 .

θ_0 [ML]	T_p [K]	Slope	Intercept	E_{des} [kJ mol ⁻¹]	ν_0 [ML s ⁻¹]
2.23	185.83	-8527.81	45.97	70.90	$1.01 \cdot 10^{19}$
3.71	176.94	-8578.10	48.09	71.32	$8.42 \cdot 10^{19}$
3.71	186.10	-3917.85	21.40	32.57	$2.15 \cdot 10^8$
4.04	185.30	-9422.91	51.96	78.34	$4.03 \cdot 10^{21}$

3.4 Monolayer desorption analysis

3.4.1 Experimental data

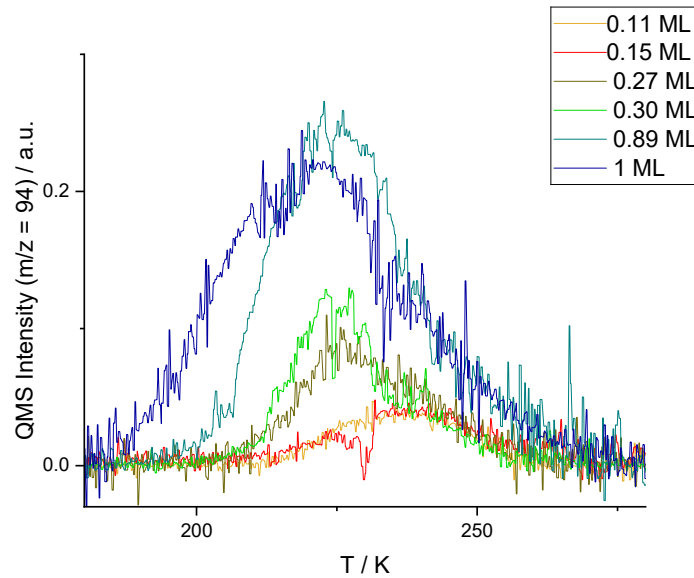


Figure 12: Thermal desorption spectra for initial coverages ranging from 0.11-1.00 ML. The QMS Intensity [a.u.] for $\frac{m}{z}=94$ u/e is plotted against the temperature.

In Figure 12 can be observed that the peak height does not strictly increase with coverage. Instead, the area under the desorption peak increases as initial coverage increases, which is confirmed by the integral values in Table 4. Qualitatively, this can be attributed to the inconsistent peak width observed for each initial coverage, especially between the spectra for 0.88 ML and 1 ML. This is most likely due to the presence of more than one binding site for phenol on an Ag (111) surface, which may have distinct bonding energies. Each binding site manifests itself as a peak on a TPD spectrum, and when the desorption energy differences are relatively similar, and the peak widths are relatively wide, the sum of these peaks results in a broader peak.

Just by inspection, it is difficult to derive any correlation between peak temperature, T_p and the initial coverage. Even then, mathematically determining the peak temperature proved to be challenging and unreliable with the raw TPD spectra because of the high amount of noise present. Peak temperature was therefore determined by first fitting a LOESS (Locally Estimated

Scatterplot Smoothing) curve over spectral data for each coverage. T_p was then determined by peak-fitting the corresponding LOESS curve. The T_p values obtained were then plotted against coverage.

As shown in Figure 13, T_p is weakly negatively correlated, poorly fitted with a linear regression, with a cumulative drop in T_p spanning 16.4 K. The drop in T_p is still not sufficient nor consistent enough for it to qualify as second order desorption kinetics, but nevertheless significant (7 % of T_p). This could be explained by the increasing electrostatic repulsion between phenol adsorbates with increasing initial coverage. At high coverages, the average distance between phenol molecules decreases, resulting in stronger repulsive interactions between phenol molecules, thus counteracting the attractive potential between phenol and the Ag(111) adsorption site. Modelling this interaction analytically will be challenging, as the dependency of T_p on initial coverage will most certainly be non-linear.

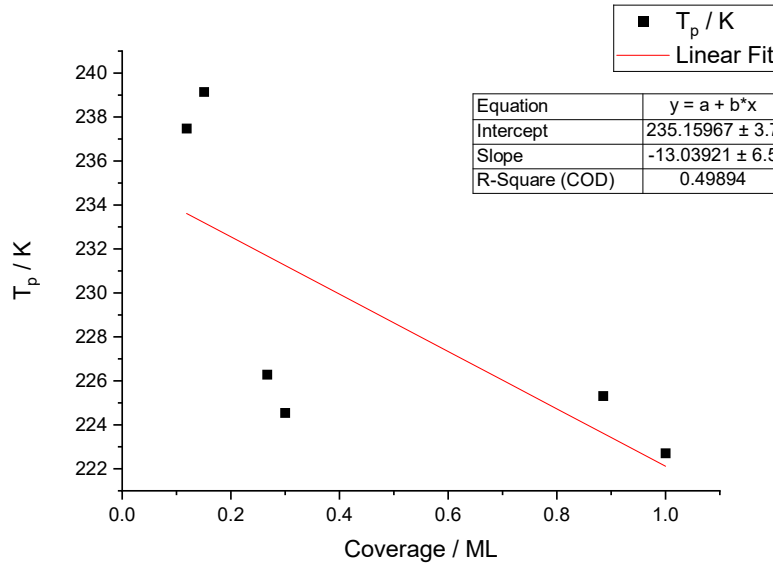


Figure 13: A scatterplot of T_p against initial coverage in monolayers. A linear fit is added into the plot, to show that the data is not following linear behaviour. Its properties are listed in the box at the top right.

For first order desorption, the relationship between E_{des} and T_p can be derived from the first-order Polanyi-Wigner equation (Equation (1)). Assuming $\ln(\frac{E_{des}}{k_B T})$ is relatively constant for the energy scales relevant to desorption, the following expression by G. Attard and C. Barnes^[2] can be used to compute E_{des} from T_p :

$$E_{des} = k_B T_p \left(\ln \left(\frac{\nu_0 T_p}{\beta} \right) - 3.64 \right), \quad (8)$$

where ν_0 is the pre-exponential factor, assumed to be $1 \cdot 10^{13}$ and β is the heating rate, in this case 1 K s^{-1} . For monolayer desorption, an E_{des} of $60.0 \pm 1.7 \text{ kJ mol}^{-1}$ was obtained, which is seen in table 7.

Table 7: First order monolayer desorption Energies E_{des} for only low coverages.

Coverage [ML]	T_p [K]	E_{des} [kJ mol ⁻¹]
0.151	239.1	63.17
0.119	237.5	62.71
0.267	226.3	59.67
0.300	224.5	59.19
0.885	225.3	59.40
1.000	222.7	58.70

3.4.2 Simulated thermal desorption spectra

In order to simulate a TPD-spectrum, an exact analytical solution for the first order Polanyi-Wigner equation, *i.e.*, desorption rate as a function of surface temperature, $r_{des}(T)$, is necessary. This is by no means trivial, as r_{des} is dependent on θ , which is its own integral, as per the following equation:

$$r_{des}(T) = \left(\int_{T_0}^T r_{des}(T) dT \right) \frac{\nu_0}{\beta} \exp \left(-\frac{E_{des}}{k_B T} \right). \quad (9)$$

An expression that solved Equation (8) was derived, which can be further solved numerically to output r_{des} given a temperature input.

$$r_{des}(T) = \theta_0 \nu_0 \exp \left(-\frac{E_{des}}{k_B T} \right) \cdot \exp \left(\frac{\nu_0 E_{des}}{k_B \beta} \left(\text{Ei} \left(-\frac{E_{des}}{k_B T} \right) + \frac{k_B T}{E_{des}} \exp \left(-\frac{E_{des}}{k_B T} \right) \right) \right). \quad (10)$$

The $\text{Ei}(x)$ function here refers to the exponential integral. Using this expression, the thermal desorption spectra for $\theta_0 = 0.11$ ML, 0.30 ML, and 1.00 ML were simulated (Figure 14). Parameters of $E_{des} = 60.0$ kJ mol⁻¹, $\nu_0 = 1 \cdot 10^{13}$ molecules cm⁻² s⁻¹, and $\beta = 1$ K s⁻¹ were used. The spectra were simulated *via* Python 3.8.

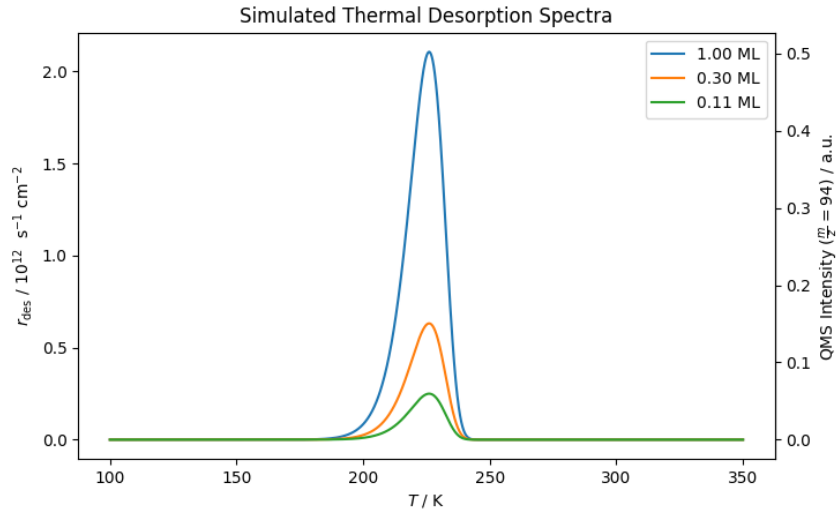


Figure 14: Simulated thermal desorption spectra of the three coverages of 1.00 ML, 0.30 ML and 0.11 ML. It shows the desorption rate r_{des} and the QMS Intensity [a.u.] for $\frac{m}{z} = 94$ u/e plotted against the temperature.

4 Discussion

4.1 Residual gas spectra

The residual gas spectra gave the results which were expected. All the important gases out of the atmosphere and water can still be found in the RG-spectra, but only in a really low intensity. This leads to the conclusion, that the chamber is mostly shut and the vacuum is stable. The only concern would be, that both, phenol and benzene, are one atomic mass unit lighter than their molecular mass would have been with calculation and what J. Lee et al. and Thomas J. Rockey et al. used in their experiments.^[3,6] This can be explained with deprotonation of the molecules. Another explanation could be a small measurement error for a larger mass to charge ratio. Nonetheless the different masses could be assigned correctly.

4.2 TPD-spectra

The received data out of the TPD-spectra looked pretty similar to the proposed data from J. Lee et al., with the biggest difference being the shift of the overall desorption temperature by 16 K compared to their experiment. The biggest differences between the two experiments would be the different adsorbing temperature and the doubled heating rate from J. Lee et al.. However these two factors should not change the experiment to this degree to get a desorption temperature difference of this height. Another reasonable thought could be a small malfunction in the not ideal cooling system of sample and chamber. The visible rise of desorption in higher temperatures underlines this idea.

On the other side, the shape of the spectra looked pretty much as expected from J. Lee et al., including form and desorption temperature differences between the desorption steps. It was not ideal, that the data-file of the multilayer-desorption at a dosis of $2.6 \cdot 10^{-7}$ mbar·s could not get analyzed, but the given data gives a reasonable good picture of how the desorption of phenol on an Ag(111)-surface should look like.^[3]

4.3 Monolayer desorption

The ideal monolayer is most likely not at the exact dosage of $1.6 \cdot 10^{-7}$ mbar·s, but it is reasonably close enough to the real value. For the exact dosage, more measurements in this dosis region would be needed. Compared to the data of J. Lee et al. it should be located in this dosage region. For finding a desorption-spectrum close to the ideal monolayer, the recorded data gives a good approach.^[3]

As demonstrated in Section 3.2.3, to obtain sticking factors values that are mostly below 1, a unit cell of $p(6 \times 6)$ had to be assumed, which is much bigger than expected. Using a overlayer unit cell of $p(2 \times 2)$ led to sticking factors much larger than one. It is, however, unusual for phenol to organise itself into a relatively big $p(6 \times 6)$ surface cell. Phenol molecules have been shown to adsorb onto Pt(111) *fcc* with a surface unit cell as small as $(\sqrt{7} \times \sqrt{7})R19.1^\circ$.^[7] Thus, the only reasonable conclusion is that the calculated impingement values, N_{imp} , are too low. This might

have been due to some inaccuracy from recording the dosages, or a faulty manometer (less likely). The large variance in sticking factors across all coverages can be explained by inconsistencies during the dosing procedure, for example, dosing at different pressures may result in different number of molecules impinging, despite a similar recorded dosage value. This can be rectified by carrying out the experiment with different dosages with otherwise identical magnitudes, like dosing the TPD experiments at the same pressure, having the same background pressure at the start of the experiment and always having the same closing delay when closing the valve.

For monolayer desorption, the E_{des} calculated was $60.0 \pm 1.7 \text{ kJ mol}^{-1}$, which is close to the range of values reported by J. Lee et al., $61.1 - 64.9 \text{ kJ mol}^{-1}$ depending on different values of ν_0 used in Equation (8) in that paper. This tiny difference was reflected in the peak temperatures obtained ($223 - 239 \text{ K}$), which are consistently lower than those in the J. Lee et al. paper (*ca.* 250 K).^[3] This 15 K gap could be due to the difference between the recorded temperature on the thermostat and the true surface temperature, which may be influenced by the chamber temperature. Regardless, this deviation is within 5% of the literature value, which is not significant.

4.4 Multilayer desorption

The obtained E_{des} for multilayer desorption is $73.0 \pm 3.9 \text{ kJ mol}^{-1}$, which overlaps the value reported by J. Lee et al., $70.7 \pm 2.5 \text{ kJ mol}^{-1}$.^[3] Since this is multilayer desorption, the main interaction is between phenol and phenol, which is quantified through the enthalpy of sublimation for phenol, $\Delta H = 67.8 \text{ kJ mol}^{-1}$. This is just slightly lower than the calculated E_{des} , because in bulk sublimation, the attractive potential of the surface is not a factor.

E_{des} of the phenol multilayer is also higher than E_{des} of the phenol monolayer, $60.0 \pm 1.7 \text{ kJ mol}^{-1}$. This has important repercussions, as it demonstrates that it is energetically more favourable for phenol multilayers to form instead of phenol adsorption onto Ag(111). Thus, for phenol, it is not the case that multilayers form exclusively after a monolayer is formed when phenol was impinged onto the surface. This makes it more challenging to determine when a full monolayer is achieved, because the multilayer would start growing even before the monolayer is saturated.

4.5 Simulated TPD spectra

Upon inspection, the simulated thermal desorption spectra for monolayer desorption broadly follow the same profile as the monolayer desorption peaks in the J. Lee et al. paper.^[3] On top of that, the simulated spectra from this experiment exactly match the simulated spectra reported by R. Gorte and LD Schmidt^[8], for first order desorption. The main difference is that peaks reported by J. Lee et al. span a significantly wider temperature range than the simulated spectra, which is also observed in the TPD spectra recorded in this experiment.^[3] This can be attributed to the presence of different binding sites for phenol on the Ag (111) surface. Referring to Figure 9, it is apparent that there are two distinct hollow sites - the *hcp* binding site and the *fcc* binding

site. Since Ag is an *fcc* lattice, these two sites differ energetically, which should result in two monolayer desorption peaks. However, if both peaks are close to each other in energy, and hence peak temperature, and are broad, the net result on a TPD spectrum will be one broader peak.